

Dynamic assessment and prediction of land use alterations influence on ecosystem service value: A pathway to environmental sustainability

Sujit Kumar Roy^{a,*}, Md. Tauhidul Alam^b, Pratik Mojumder^c, Ismail Mondal^d, Abdulla - Al Kafy^e, Mithun Dutta^a, Md Nahid Ferdous^f, Md. Abdullah Al Mamun^g, Sania Binte Mahtab^h

^a Institute of Water and Flood Management, Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

^b Department of Agriculture, Noakhali Science and Technology University, Bangladesh

^c Dept. of Environmental Science and Disaster Management, Daffodil International University, Bangladesh

^d Department of Marine Science, University of Calcutta, Kolkata, West Bengal, 700019, India

^e Department of Geography and the Environment, The University of Texas at Austin, 1 University Station A3100, Austin, TX, 78712, USA

^f Institute of Disaster Management, Khulna University of Engineering & Technology, Bangladesh

^g Agricultural Statistics Division, Bangladesh Rice Research Institute, Bangladesh

^h Department of Water Resources Engineering, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh

ARTICLE INFO

Keywords:

Urbanization
Green cover
Ecosystem service value
Google earth engine
Sustainable environment

ABSTRACT

In the face of rapid urbanization, understanding and managing changes in Land Use and Land Cover (LULC) is crucial for the sustainability and resilience of urban ecosystems. Rapid urban growth resulting from changes in LULC, can lead to the fragmentation and degradation of Ecosystem Services (ES), including habitat provision, water regulation, and air purification. This study investigates LULC changes in Dhaka City, Bangladesh, from 2010 to 2021 and forecasts their future impacts on ES in 2030 and 2040. Landsat images were processed using the Google Earth Engine platform, and a cellular automata technique was integrated with an Artificial Neural Network (ANN) for LULC mapping and prediction. Results showed an increase in built-up areas from 31% to 42% between 2010 and 2021, with a projected decrease to 34% and 32% in 2030 and 2040, respectively. Water bodies, vegetation, and agricultural land declined from 2010 to 2021 but are expected to increase in 2030 and 2040. This trend indicates a growing awareness of the need for a sustainable environment and resilience in urban ecosystem development. The value of ES is predicted to drop from \$172.66 to \$160.23 over thirty years (2010–2040), primarily due to the degradation of water bodies (\$7.64) and agricultural land (\$6.89). However, the projected increase in water bodies, vegetation, and agricultural land by 2040 suggests a potential for recovery and enhancement of these ES values through sustainable practices. Future land use policies should promote sustainable agriculture and water resource management while fostering resilience against rapid urban growth. This approach will ensure the sustainable development of urban ecosystems. This study informs future land use strategies in Dhaka City, supporting sustainable natural resource management, urban ecosystem quality management decisions, and resilience planning. It underscores the importance of integrating sustainability and resilience into urban development policies for the benefit of both the environment and the city's inhabitants.

1. Introduction

Ecosystems are the unsung heroes of the planet, providing essential services that are fundamental to human health, livelihoods, and ultimately, survival (Past et al., 2022). The concept of Ecosystem Services (ES) has been gaining traction in both scientific and public policy arenas

(Khwarahm, 2021), shedding light on how natural ecosystems and their resident species bolster human life and enhance living conditions. The United Nations' Millennium Ecosystem Assessment (MEA) in 2005 marked a significant milestone in the recognition of ES. This four-year research project, involving 1300 scientists, was a clarion call to policymakers worldwide. Hot on its heels was the Economics of Ecosystems

* Corresponding author. Institute of Water and Flood Management, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh.

E-mail addresses: sujitroy.bejoy@gmail.com (S.K. Roy), mdtauhid481@gmail.com (Md.T. Alam), pratikmojumdar@gmail.com (P. Mojumder), ismailmondal58@gmail.com (I. Mondal), abdullaalkafy@utexas.edu, abdulla-al.kafy@localpathways.org (A.A. Kafy), mithundm23@gmail.com (M. Dutta), nahidferdous103@gmail.com (M.N. Ferdous), mamunru4777@gmail.com (Md.A. Al Mamun), [saniam@wre.buet.ac.bd](mailto:sania@wre.buet.ac.bd) (S.B. Mahtab).

<https://doi.org/10.1016/j.indic.2023.100319>

Received 13 July 2023; Received in revised form 21 November 2023; Accepted 23 November 2023

Available online 5 December 2023

2665-9727/© 2023 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

and Biodiversity (TEEB) initiative by the UN Environment Programme between 2007 and 2010 (TEEB Foundations, 2010). ES can be categorized into four broad types: (a) provisioning services like food production, (b) regulating services such as climate regulation, (c) supporting services including pollination, and (d) cultural services like recreation. These services offer a cornucopia of benefits that enrich human life (Khwarahm et al., 2021). Costanza (Past et al., 2022) identified 17 functions derived from ES that are instrumental to human survival and well-being. It's like nature's own Swiss Army knife, equipped with tools to support every aspect of human existence. Today, businesses and the general public are waking up to the value of ES. The Global Business Council for Sustainable Development is leading the charge in promoting this concept (Past et al., 2022), (Khwarahm, 2021), (Khwarahm et al., 2021). It's clear that when it comes to preserving the planet and ensuring human survival, ES isn't just part of the conversation its the conversation.

Urban ES play a crucial role in connecting cities with the biosphere, reducing the ecological footprint and debt of cities, enhancing urban resilience and quality of life, and promoting public health ("MOLUSCE). They perform a variety of functions, including food and water provision, waste management, regulation of the urban heat island effect, water purification, noise reduction, pollination facilitation, and climate regulation. These functions directly or indirectly contribute to human well-being (Tomar et al., 2021), ("MOLUSCE), (Li and Li, 2015), (Saha et al., 2021) However, human activities have significantly and rapidly impacted ecosystems over the past 50 years, more than any comparable period in human history (Baig et al., 2022). Among these activities, changes in Land Use and Land Cover (LULC) have immediate effects on ecosystem health and integrity, which subsequently influence the availability of ES (Saputra and Lee, 2019), (Aneesha Satya et al., 2020), (Sanderson, 2010). Land cover reflects the interaction of various human and environmental factors within an urban region (Verma et al., 2020). In scientific literature, "land cover" and "land use" are often used interchangeably to refer to the LULC concept (Costanza et al., 1997). Urbanization and suburban sprawl have accelerated transitions in LULC (Tolessa et al., 2017). Land cover refers to the natural ecological covering of Earth's surface, including vegetation, water, soil, and other natural surfaces. In contrast, land use describes how landscapes in urban and rural settings have been altered by humans (Ongsomwang et al., 2019). The spatial distribution of LULC within an urban area influences the ES provided by the urban environment. For instance, transforming a wetland into a built-up area for human habitation can lead to increased urban flooding due to heavy precipitation and reduced land water absorption capacity ("MOLUSCE), (Tomar et al., 2021), (Li and Li, 2015), (Saha et al., 2021), (Baig et al., 2022). This can intensify waterlogging, damage infrastructure, and disrupt city residents' daily lives. Additionally, the loss of wetlands may negatively affect local biodiversity and reduce the ecosystem's ability to provide essential services such as water purification, carbon sequestration, and climate regulation (Saputra and Lee, 2019), (Aneesha Satya et al., 2020).

Bangladesh, one of the world's most disaster-prone countries, is highly susceptible to climate change. Its agroecomic system is heavily reliant on emerging countries, and it has a dense population. As the century turned, its economy experienced a boom, making it one of the most dynamic in Asia and globally (Pullanikkatil et al., 2016). Numerous studies have shown that Dhaka, the capital of Bangladesh, has seen significant changes in LULC in recent decades. These changes are due to rapid urban expansion, economic growth, climate change, rising sea levels, and natural disasters (Kreuter et al., 2001), (Birhane et al., 2019). Except for 1990, built-up areas have dominated all types of LULC. Between 1990 and 2020, the built-up area increased by 188.35%, growing at an average rate of approximately 6.28% per year (Pullanikkatil et al., 2016; Kreuter et al., 2001; Birhane et al., 2019). This rapid urbanization, coupled with population mobility, industrialization, and economic expansion, resulted in Dhaka's built-up area expanding at the expense of water bodies, agricultural land, forests, and vegetation.

This transformation of the urban area led to a reduction in the ecosystem service value (ESV) by 59.55% (\$85 million), from \$142.72 million in 1990 to \$57.72 million in 2020. According to elasticity test findings, for every one percent change in LULC during the investigated period, there was a 0.33% increase or decrease in total ESV. However, the most significant decline in total ESV occurred between 1990 and 1995, amounting to 32.20%, or \$27.32 million. Water bodies had a stronger positive impact on ESV change compared to other LULC categories.

Moreover, studies have shown that water supply, waste treatment, and disturbance regulation are the primary factors contributing to ESV (Naikoo et al., 2020). There are several software programs available for simulating future LULC scenarios, including LCM (Land Change Modeler), CA-MARKOV, DINAMICA-EGO, and CLUE-S, which employ empirical approaches based on previous LULC (Past et al., 2022), (Khwarahm, 2021), (Khwarahm et al., 2021). The MOLUSCE plugin in QGIS is now accessible for evaluating current LULC changes and projecting future LULC. This module employs the Markovian approach to create a transition potential matrix and trains a simulation model using Multi-Criteria Evaluation (MCE), Artificial Neural Networks (ANN), Logistic Regression (LR), and Weights of Evidence (WoE). Monte Carlo cellular automata modeling is then used to produce the simulated land use map ("MOLUSCE), (Tomar et al., 2021), (Li and Li, 2015) Recently, researchers have successfully combined CA models with ANN to predict future LULC patterns globally (Saha et al., 2021), (Baig et al., 2022) Hybrid CA-ANN models provide numerous advantages over traditional CA-MC models, including improved precision, the ability to capture non-linear relationships, efficient handling of large datasets, greater flexibility, and increased prediction reliability (Saputra and Lee, 2019), (Aneesha Satya et al., 2020). CA is a method used to simulate the behavior of individual units that interact based on predefined rules. It's particularly useful in simulating changes in land cover over time and evaluating land management strategies (Chen et al., 2016), (Wu, 2002). Machine Learning (ML), on the other hand, employs algorithms to identify patterns in data and make predictions. It has various applications in Ecosystem Services (ES) studies, including classification, regression, clustering, and anomaly detection. ML can classify satellite images or predict changes in land cover based on environmental factors. The choice between CA and ML depends on research needs, data availability, and resources. ANN-CA is often considered the most powerful tool for predicting spatiotemporal changes in land cover (Saputra and Lee, 2019), (Past et al., 2022), (Rahman and Rahman, 2023), (Pandey and Kumari, 2023), (Nahrin, 2020), (Rahman et al., 2019), (Zinia and McShane, 2018).

Understanding land cover dynamics and their impacts on Dhaka's ES is crucial. For instance, Dhaka has significantly expanded its built-up area, often at the expense of wetlands. This transformation has altered the natural landscape, leading to issues such as flooding from rainfall and waterlogging, posing severe challenges for urban dwellers. Dhaka's current air and water pollution problems are also a result of its endangered urban ecosystem (Nahrin, 2020), (Rahman et al., 2019), (Zinia and McShane, 2018). The primary objectives of this research are to predict the dynamics of changes in land cover, estimate and map the total ESV across the study period in Dhaka City, quantify the value of each ES function, and assess the impact of these changes on ESV by determining the elasticity of ESV in response to these changes. A global dataset for ES values of various land uses was used as input. Satellite images were employed to assess the dynamics of land cover changes in the study area and quantify changes in ES value. Assessments of urban ES at regional and global levels infrequently use changes in land cover obtained from satellite images. While several studies have focused on the valuation of ES and changes in land cover in Bangladesh, more research is needed to examine how these changes impact urban ES. By understanding how changes in land cover affect ESV, local benefits can be optimized. This study contributes to existing literature by comprehensively analyzing relationships between changes in land cover, urban ES, and their economic implications. These findings can inform

policy-making and sustainable urban planning.

The novelty of this work lies in its integrated approach to assessing the impacts of changes in land cover on urban ES by combining advanced remote sensing techniques, machine learning algorithms, and economic valuation methods. This unique combination enables a more accurate and detailed understanding of the complex relationships between changes in land cover and ES in rapidly urbanizing areas like Dhaka City. The intellectual merit of this study is evident in its rigorous methodology and the development of a robust framework for predicting dynamics of land cover and estimating ESV. This framework can be applied to other urban contexts and ecosystems, providing valuable insights for researchers, policymakers, and planners in urban ecology, environmental management, and sustainable development. The broader impacts of this manuscript are manifold. It contributes to the growing body of knowledge on the interplay between changes in land cover, urban ES, and human well-being, ultimately promoting a more holistic understanding of urban environments. The results of this study can inform land-use planning and policy-making, guiding urban development towards more sustainable pathways that enhance ES and improve the overall quality of life for urban dwellers. The research also has the potential to foster public awareness of the importance of ES and their role in enhancing cities' resilience against climate change and other environmental challenges, thus encouraging greater community engagement in the conservation and restoration of urban ecosystems.

2. Materials and methods

2.1. Study area

Dhaka, the capital city of Bangladesh, is nestled along the banks of the Buriganga River, situated between latitudes $23^{\circ}40'$ and $23^{\circ}51'N$ and longitudes $90^{\circ}19'30''$ and $90^{\circ}30'30''E$ (Fig. 1). The city is perched on a sandy Pleistocene terrace, commonly referred to as the Modhupur

terrace (Dasgupta et al., 2018). The city spans an area of 300 km^2 and is elevated between 6.5 and 9 m above mean sea level. Dhaka is surrounded by various bodies of water, including the Balu River to the east, Turag River to the west, and Buriganga River to the south and southeast. These natural features contribute to the city's unique ecosystem and biodiversity. The city's climate is tropical, characterized by heat, humidity, and significant rainfall. According to the Bangladesh Meteorological Department, average annual rainfall of Dhaka is 2500 mm, with about 80% falling between July and August. In addition, the average temperature varies throughout the year is 16°C – 36°C .

Dhaka is an ideal study area for urban ES due to its rapid urbanization and diverse ecosystems. The city has experienced significant changes in LULC due to economic growth, climate change, rising sea levels, and natural disasters (Dewan and Yamaguchi, 2009). These changes have reduced ES, impacting the well-being of urban communities (Yin et al., 2021). Moreover, Dhaka's unique geographical location and climate make it a vibrant hub of biodiversity. Its dynamic weather patterns and diverse ecosystems offer a rich tapestry of natural phenomena waiting to be explored. Studying Dhaka provides valuable insights into the impacts of land use changes on urban ES. The findings can inform policy-making and sustainable urban planning, contributing to its residents' well-being and preserving its biodiversity (Dewan and Yamaguchi, 2009), (Yin et al., 2021), (Rahaman et al., 2023).

2.2. Data description

This research utilized a combination of primary and secondary data, including elevation, slope, aspect, distance from the river, and distance from the road. Additionally, LULC maps for the years 2010, 2015, and 2021 were used (as shown in Table 1). Google Earth Engine (GEE) was employed for quick analysis, leveraging Google's computational infrastructure for near-real-time access to web datasets. This platform enables planetary-scale geospatial analysis for everyone. For this study

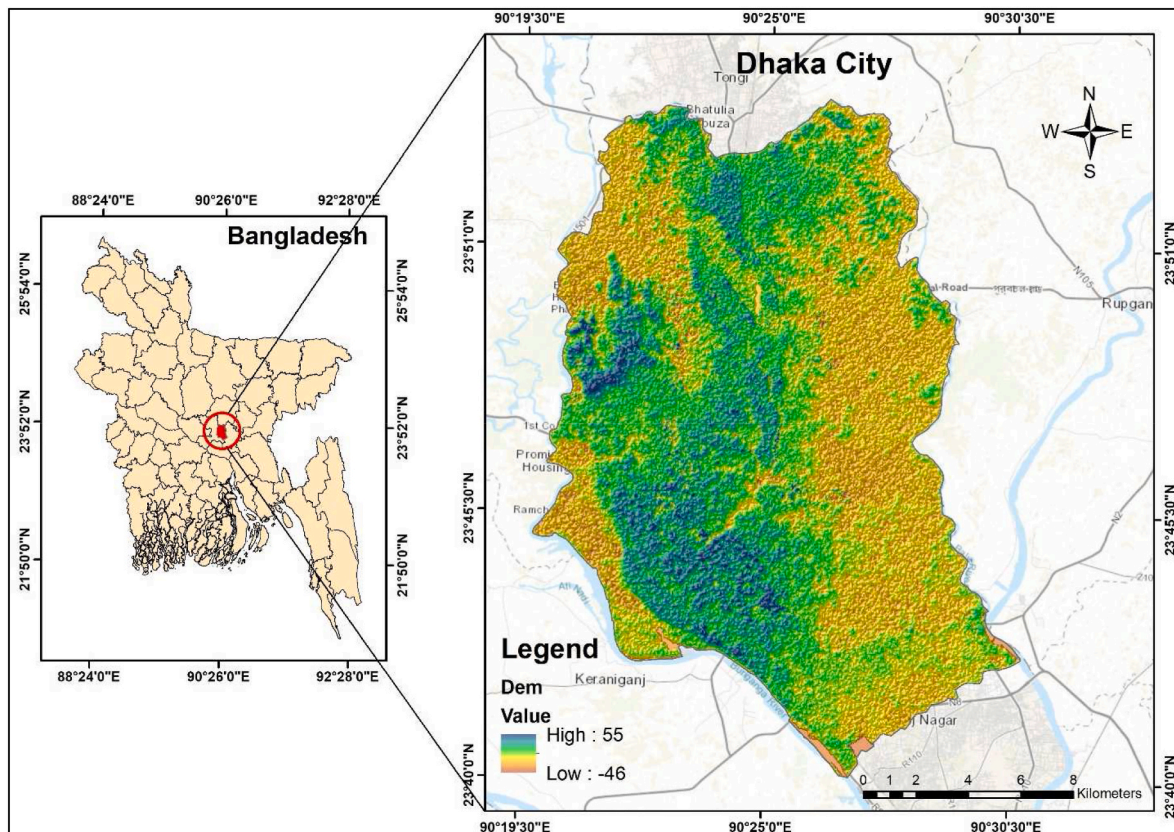


Fig. 1. Location map of the study area.

Table 1
Description of the datasets.

Data	Spatial Resolution (meter)	Year	Description	Path	Row	Cloud Cover	Acquisition Date	Source
LULC	30	2010,2015,2020,2021	Landsat (TM and OLI)	137	43	Less than 1%	September 25, 2022	https://earthengine.google.com/
Elevation		2014	DEM SRTM					https://earthexplorer.usgs.gov/
Slope								https://earthexplorer.usgs.gov/
Aspect								https://earthexplorer.usgs.gov/
Distance from river			Major and minor roads					https://www.hydrosheds.org/
Distance from road								https://extract.bbbike.org/

Table 2
Class descriptions of LULC classification.

Land use class	Description
Built-up Area	Residential areas, industrial land, transportation, communications, mixed urban, and other urban areas
Waterbody	Wetlands, inland water bodies, rivers, reservoirs, lakes, low-lying areas
Vegetation	Deciduous forest land, evergreen forest land, mixed forest land, grassland, orchard, and urban greening land
Agricultural Land	Cultivated land, fallow land, and terraced
Bare Land	No vegetation, exposed soils, landfill sites, areas of active excavation and sandy areas

region, Landsat 5 TM and 8 OLI surface reflectance images (for the years 2010, 2015, and 2021) were obtained via GEE (Table 1). GEE's Surface Reflectance images have already accounted for atmospheric influences such as aerosol scattering and thin clouds. This makes it easier to detect and characterize surface changes. The Landsat TM and OLI bands were used for classification on the GEE platform. Digital Elevation Model (DEM) data, acquired from the USGS, was used to create the elevation, slope, and aspect maps. Data used for calculating distances from roads and rivers were collected from the HydroSHEDS and BBBike websites, respectively. Furthermore, population data was obtained from the DIVA-GIS website. This comprehensive dataset provided a robust foundation for analyzing the impacts of land use changes on urban ES.

2.3. Preprocessing

Landsat optical satellite imagery, spanning spectrum values from the visible to the short-wave infrared, has been widely used to map various Earth features, including LULC, and for ecological monitoring. One of the primary challenges of using Landsat data is dealing with cloud cover that often obscures the images. This is particularly problematic in large areas and tropical climates like Bangladesh, which experience constant cloud cover throughout the year. To mitigate this issue, we intended to use GEE cloud filter. This tool helps reduce the impact of cloud cover on time series satellite remote sensing analyses. The vector shapefile of Dhaka city was used to sort and extract the mean pixel values for each year (2010, 2015, and 2021). Additionally, population data was downscaled to a geographical resolution of 30 m. This approach ensures a more precise representation of population distribution across the study area.

2.4. LULC mapping, accuracy assessment, and change detection

2.4.1. LULC classification

For LULC classification, accurate reference data and an appropriate classification system are crucial. In this study, LULC was classified into five categories: built-up areas, water bodies, vegetation, agricultural land, and barren land (Table 2). Training and testing samples were collected using Google Earth images, the Copernicus Global Land Cover Layer (CGLS-100), the L8 RGB and composite infrared layers, and spectral indices. A total of 2913 sample points were gathered for various classes in 2010, 2015, and 2021. For each class, at least 70 samples were collected; 80% of these were randomly selected as training samples for classifying the Landsat images, while the remaining 20% served as validation samples to verify the classification accuracy. Several supervised classification packages are available in GEE, including Random Forests (RF), Classification and Regression Trees, Support Vector Machines (SVM), and Naive Bayes. The RF classifier has gained popularity over the past two decades due to its high classification results and processing speed. RF is an ensemble technique that can enhance classification performance and assess variable importance. In RF, a substantial number of trees may be present. Approximately two-thirds of the data (in-bag samples) are used to train the trees, while the remaining one-third (out-of-bag samples) is used in an internal cross-validation process to determine how well the RF model performs. The use of numerous trees makes RF less prone to errors compared to regular decision trees. As a result, in this study, an RF model was employed to generate classification maps.

2.4.2. Accuracy assessment of LULC classification

The precision with which the LULC maps are produced is directly proportional to the accuracy with which the classified LULC maps are assessed. Moreover, it shows the effectiveness of the categorization method. Accuracy in remote sensing image classification is often evaluated using the confusion matrix. It demonstrates the accuracy between the verified data and the LULC classification results. In this research, classification accuracy is assessed using the following metrics: overall accuracy, producer accuracy, user accuracy, overall accuracy, and kappa coefficient. According to Congalton, the LULC classification is satisfactory, with an overall accuracy of more than 70% and a kappa value between 0.40 and 0.85.

Overall accuracy is a standard measure of classification accuracy that calculates the proportion of correctly categorized samples by dividing the sum of the diagonal entries by the total number of pixels under study (Sanderson, 2010).

It can be calculated using the following equation: 1:

$$\text{Overall accuracy} = \frac{\text{Total number of corrected classified pixels (diagonal)}}{\text{total number of reference pixels}} \times 100 \dots \dots \dots (1)$$

In contrast to overall accuracy, user and producer accuracy gives insight into each class's classification quality. First, the producer's accuracy is calculated by distributing the total number of pixels collected from the reference data by the number of accurate pixels in one classification. In the meantime, the user's accuracy is calculated by distributing the total number of pixels classified in each class by the number of pixels that have been correctly recognized in that class.

The following Eqs. (2) and (3) is used to calculate the producer's and user accuracy:

$$\text{Producer Accuracy} = \frac{\text{number of correctly classified pixels in each category (diagonal)}}{\text{total number of reference pixels in each category (column total)}} \times 100 \dots \dots \quad (2)$$

$$\text{User Accuracy} = \frac{\text{number of correctly classified pixels in each category (diagonal)}}{\text{total number of reference pixels in each category (row total)}} \times 100 \dots \dots \dots \quad (3)$$

The kappa coefficient of agreement (KAPPA) is another indicator used in error matrix analysis and often used to assess LULC classifications acquired from remote sensing and to summarize the findings of a proper investigation (Verma et al., 2020). Basically, Kappa is assigned a number between 0 and 1. The value 1 means perfect accuracy, and as it gets closer to 0, it gets less accurate. The kappa coefficient value has been evaluated differently by many researchers. The suggested threshold for kappa is 0.40; values between 0.40 and 0.55 indicate accurate results; values between 0.55 and 0.70 indicate excellent accuracy; values between 0.70 and 0.85 specify perfect accuracy; and values more than 0.85 represent the maximum accuracy between different images.

The following Eq. (4) is used to calculate the kappa coefficient:

$$\text{Kappa Coefficient (T)} = \frac{\text{Total number of Sample} \times \text{Total Number of Corrected Sample} - \sum (\text{Column Total} \times \text{Row Total})}{(\text{Total number of Sample})^2 - \sum (\text{Column Total} \times \text{Row Total})} \times 100 \dots \dots \dots \quad (4)$$

2.5. Change detection of LULC

In this study, the process of change detection was carried out to analyze shifts in various LULC classes. This analysis resulted in the creation of three distinct LULC change maps. These maps provide a visual representation of how land use and land cover have evolved over specific periods: 2010 to 2015, 2015 to 2021, and a comprehensive view from 2010 to 2021. The change detection process was performed using the MOLUSCE plugin in QGIS, a powerful tool for analyzing and visu-

alizing geospatial data ("MOLUSCE., 2014), (Ongsomwang et al., 2019), (Pullanikkatil et al., 2016). This plugin allows for the comparison of LULC data from different time periods, highlighting areas of change. The resulting change maps display areas that have remained under a similar LULC class during these periods, indicating regions with stable land use practices. They also illustrate the transition from one LULC type to another, shedding light on the patterns and pace of urban development or natural resource depletion. This comprehensive approach provides valuable insights into the dynamics of LULC over time. It offers a detailed understanding of how these changes impact local ecosystems and can inform sustainable land management strategies ("MOLUSCE), (Ongsomwang et al., 2019).

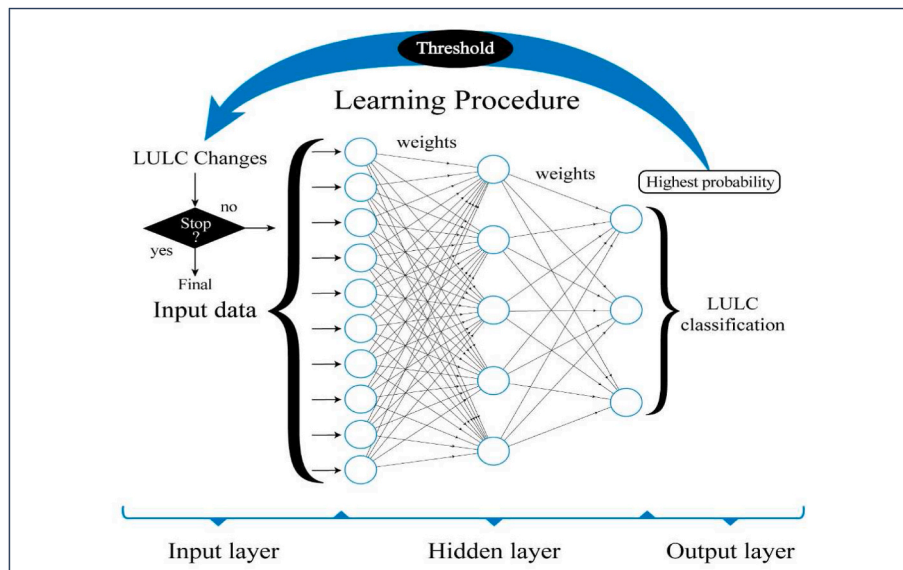


Fig. 2. Structure of ANN-CA model for modelling LULC changes.

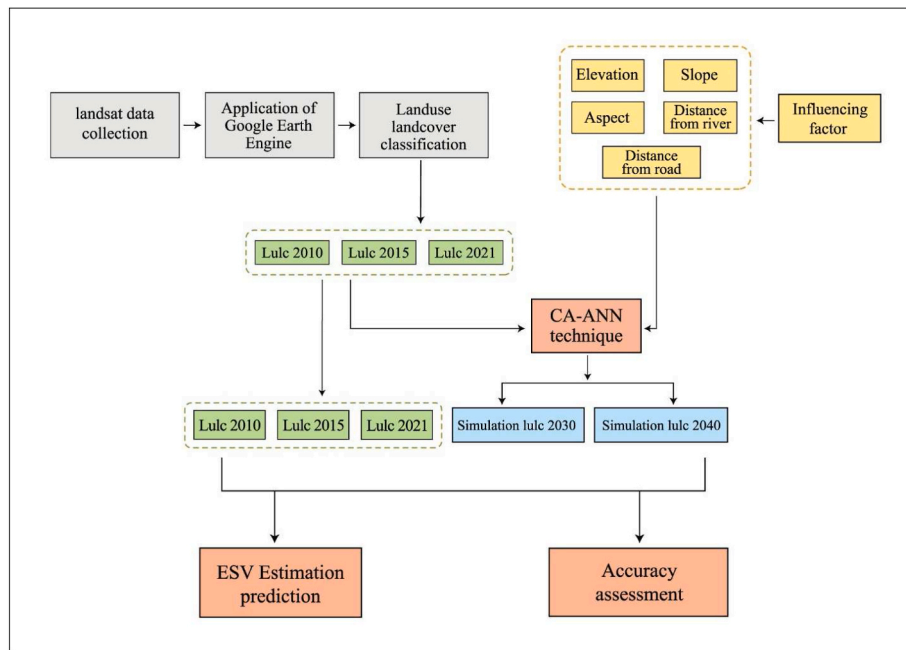


Fig. 3. Methodological flowchart.

2.6. ANN-CA modeling

Over the past two decades, numerous simulation-based models have been developed and are currently being utilized to simulate changes in land cover globally. This research employs CA-ANN techniques to simulate and predict LULC changes (Costanza et al., 1997). ANNs have gained significant popularity in the field of remote sensing for effective LULC modeling and classifications in recent years (Tolessa et al., 2017). An ANN is composed of neurons, similar to those in the human brain, and leverages data to identify trends. The ANN model used in this study is depicted in Fig. 2. Changes in LULC are represented through CA, using transition probabilities derived from the ANN training. The ANN employed the multilayer Perceptron method with a learning rate of 0.100, a 1-pixel neighborhood, a momentum of 0.050, and underwent 1000 iterations across 10 hidden layers. The ANN-CA modeling was executed using QGIS and its MOLUSCE plugin (Ongsomwang et al., 2019), (Pullanikkatil et al., 2016), following a step-by-step process. This approach provides a comprehensive framework for understanding and predicting changes in LULC (see Fig. 3).

Step 1:

The first phase defines the neural network's input layers for the LULC simulation. Then, the simulation is performed at the pixel level, with each pixel receiving n-attributes as input to the neural network. Finally, Eq (5) describes the attributes of a neural network.

$$X = [x_1, x_2, x_3, \dots, x_n]^T \dots \dots \dots (5)$$

where, $x_1, x_2, x_3, \dots, x_n$ denotes to the attributes and T denotes to transposition.

The initial (2010) and final (2015) LULC maps and five spatial parameters were used as the input layer. All layers were rescaled in raster (.tiff) format with the same spatial resolution (30 m) and then processed to obtain the attributes.

Step 2:

The second stage involves determining the correlation between the variables using a two-way raster assessment, where the 1st raster is

derived from one variable and the second is derived from another. Then, between the initial (2010) and final (2015) periods, the LULC area and changes are calculated for each class. Moreover, a transition matrix shows the percentages of pixels that flip between various categories.

Step 3:

At this stage, ANN was used to predict the transition probability. There are three levels to a neural network: input, hidden input, and hidden output. After scaling in the [0, 1] range, each spatial parameter is linked to a neuron as an input layer. The 10 neurons that match the 10 attributes are used as an input layer as a function. The message sent to the k th neuron by the j th neuron from the input layer at time t was computed in the hidden layer using the notation $net_j(k, t)$ in Eq. (6).

$$net_j(k, t) = \sum_i w_{ik} \hat{x}_i(k, t) \dots \dots \dots (6)$$

Where w_{ik} denotes the weight between the input and hidden layers and $\hat{x}_i(k, t)$ denotes the i th scaled attribute connected to the i th neuron as an input layer in relation to the k th cell at time t .

In order to overcome challenges with classification, the output layer of the application neural network must contain the same number of neurons as the predicted classes of the target or dependent variable. As a result, the output layer is comprised of the six neurons corresponding to the five LULC classes. In addition, the neuron l contained in the output layer is responsible for producing a value that denotes the transition probability from the basic LULC type l . The output function of the neural network, the transition probability, is determined using equation (7).

$$P(k, t, l) = \sum_j w_{jl} \frac{1}{1 + e^{-net_j(k, t)}} \dots \dots \dots (7)$$

Hence, $P(k, t, l)$ denotes the potential of converting to l -class LULC from the existing at time t for the cell k , whereas w_{jl} refers to the weight as the hidden and output layers.

The momentum and learning rates used to train the potential transition model in this study were 0.050 and 0.100, respectively. Moreover, a neighborhood value of 1 pixel with 10 hidden layers was chosen, and the number of iterations was increased to 1000 to avoid the problem of overfitting in the model.

Step 5:

In this step, the simulated map 2021 was compared with the classified map 2021 to verify the ANN model and prediction accuracy. The overall kappa coefficient was computed in the range of 0–1. The high level of agreement and acceptable kappa coefficient value demonstrated that the simulation mode had been validated.

Step 6

LULC maps for 2030 and 2040 were generated after validation to assume that the dynamics and existing patterns of the LULC change will continue.

2.7. Assessment of ESV

The estimate of ES has attracted much interest, and it is now being utilized worldwide to comprehend the many advantages of ESV better. In the last several decades, many research was carried out to determine the value of ES, such as the economic evaluation of tropical forests, the management of endangered species, and protected areas. Nonetheless, the Costanza et al. (1997) ecosystem service valuation model is regarded as the most reliable way to calculate the economic value of ES and to calculate ESV using data from 16 biomes and 17 ES. Based on literature and calculations, they estimated worldwide economic values for ES using the benefits transfer technique (BTM). Since that day, this methodology has been used in many research projects evaluating ES worldwide. The Costanza et al. (1997) technique was criticized for its low estimate of cropland unit value and uncertainty, but it offers a complete set of beginning assumptions for valuing ES. Consequently, we employed this methodology in our research to value ES for Dhaka. We matched the five LULC classifications with the 16 biomes established by Costanza et al. (1997) to determine the ES values for each of the five LULC classes in Dhaka. The biome represented the LULC type with the highest proportion of relevance for that particular category.

For example, the biomes "cropland" for "agricultural land," "wetland" for "water body," "tropical forest" for "forest and vegetation," "urban" for "built-up region," and "desert" for "bare land" were allocated for the proxy to compute the value of ES for each LULC type. In this particular instance, most of the waterbodies are comprised of wetland areas, and just a minor percentage are shallow lakes and rivers. Thus, we designated wetlands rather than lakes or rivers. Several studies have used LULC categories and corresponding biomes, even if they do not match (Tolessa et al., 2017), (Pullanikkatil et al., 2016). Table 3 shows the LULC kinds, comparable biomes, and related value coefficients.

2.8. Calculation of ESV

This research followed (Costanza et al., 1997) 's formula for assessing ESVs of different LULC patterns as follows. Equation (8) multiplied the land area by the value coefficient to get the total value of ESV based on the coefficient value in Table 3.

$$ESV = \sum (A_k \times VC_k) \dots \dots \dots (8)$$

Table 3
Land use categories equivalents for biomes identified in literatures.

LULC Types	Equivalent biome	ES coefficient (US\$ ha ⁻¹ per year)
Agricultural land	Cropland	5567
Water body	Wetland	12512
Vegetation and Forest cover	Tropical Forest	5328
Built-up area	Urban	6661
Bare land	Desert	0

Whereas ESV is the overall estimated value of ES, A_k stands for the area in hectares, and VC_k stands for the value coefficient in US dollars per year per LULC category 'k'. We evaluated the variations in predicted ESV for each LULC classification (1990, 1995, 2000, 2005, 2010, 2015, and 2020) to assess the change in ESV. Equation (9) was used to calculate the ESV change rate.

$$ESV_{cr} = \frac{ESV_j - ESV_i}{ESV_i} \times \frac{1}{T} \times 100\% \dots \dots \dots (9)$$

Where T is the research period, ESV_i and ESV_j are the starting and final ES values, and ESV_{cr} is the annual ESV change rate for a particular LULC type.

2.9. Coefficient of sensitivity (CS)

In this technique of land value, several biomes were utilized as stand-ins for the many types of land uses, although the two categories may need to correspond appropriately. In particular, there are questions about aquaculture's contribution to ES. Therefore, the overall ES of the study area could be overestimated or underestimated depending on how this land-use class is estimated. Because of the imprecision of the proxy values, an elasticity analysis was performed, which measures how much an output changes for any given change in an input. This analysis yielded a sensitivity coefficient. For each of the following categories: agriculture, forests, aquaculture, rivers, settlements, and others, the ecosystem value coefficients were modified by around 50%. We computed CS to determine the stability of agricultural land, rivers, and forests, matching Costanza et al.'s proxy biomes (1997) and Costanza et al. (2014). To calculate CS, the following Eq. (10) was used:

$$CS = \frac{ESV_j - ESV_i / ESV_i}{(VC_{jk} - VC_{ik}) / VC_{ik}} \dots \dots \dots (10)$$

Here, ESV is for estimated ES value, VC stands for value coefficient, 'i' and 'j' stand for original and modified values, correspondingly, and 'k' stands for LULC class. The CS is the relation between the percentage increase or decrease in the estimated total ESV and the percentage increase or decrease in the corrected valuation coefficient (VC). A ratio of one or more suggests an elastic estimate of ES, whereas a ratio of one or less shows an inelastic and robust assessment of ES. An appropriate ecological value coefficient is often used more critically when a more significant proportionate change in the ES is matched to comparative changes in the VC (Kreuter et al., 2001).

3. Result and discussion

3.1. LULC change analysis during 2010–2040

The distribution of LULC classification categories in Dhaka is shown in Table 4. During the previous 11 years, there was a significant decrease in the vegetation cover area and an increase in the built-up areas. As a result, water bodies, forests, vegetation, and agricultural land have declined over the last three decades (Fig. 4). Furthermore, the built-up

Table 4
LULC area changes in Dhaka City from 2010 to 2021.

LULC Class	2010		2015		2021	
	Area (Km ²)	Area (%)	Area (Km ²)	Area (%)	Area (Km ²)	Area (%)
Built-up Area	92.528	31	105.616	35	127.514	42
Waterbody	21.340	7	16.709	6	10.850	4
Vegetation	99.755	33	75.147	25	43.199	14
Agricultural Land	56.007	19	48.640	16	44.361	15
Bare Land	31.051	10	54.570	18	74.758	25
Total	300.68	100	300.68	100	300.68	100

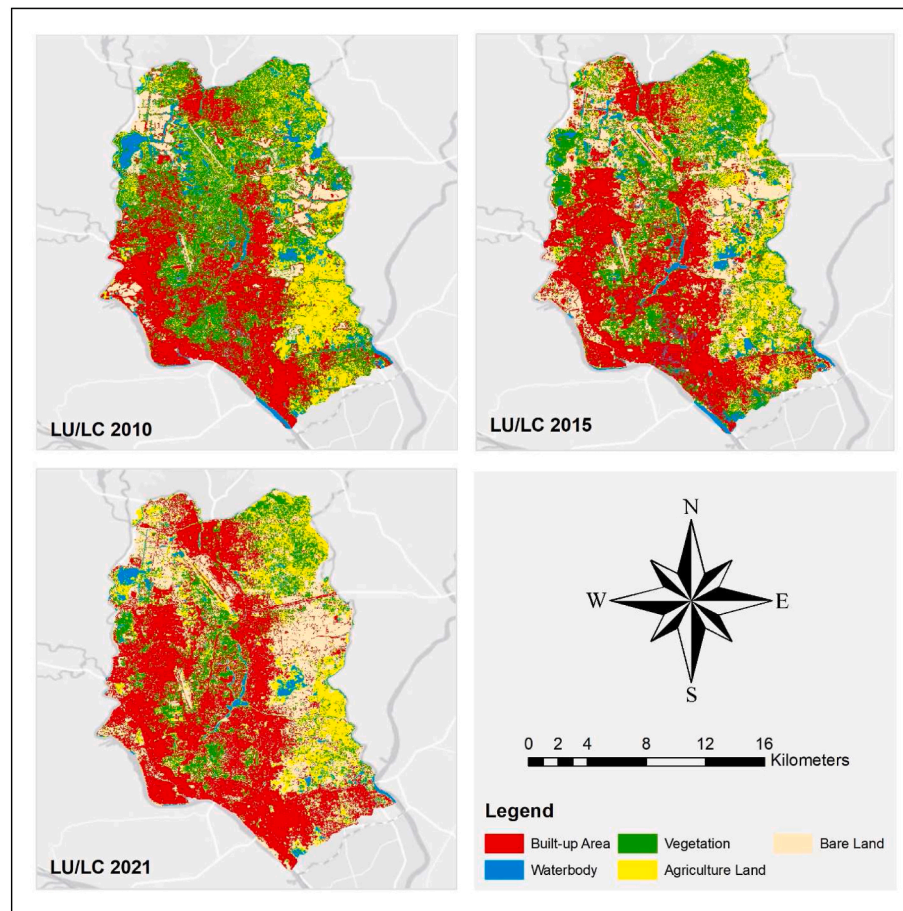


Fig. 4. LULC scenario of Dhaka city in 2010, 2015 and 2021.

and bare land has gradually increased. In 2010, the dominant LULC type was vegetation (92.528 km^2), with waterbodies covering a minor portion of land (21.34 km^2). However, in 2021, most land was dedicated to built-up areas (127.51 km^2), with waterbodies covering the most miniature land (10.85 km^2).

However, except for 2010, the built-up area was dominant in the LULC category up to 2021, increasing by about 31%, 35%, and 42% in 2010, 2015, and 2021, respectively. The most visible difference is found in the vegetation and built-up areas. The land use category's vegetation coverage was 99.75 km^2 in 2010, but it rapidly decreased and is projected to be 43.19 km^2 by 2021. Given the appropriate socio-economic setting of this location, the settlement quickly emerged as a dominant LULC due to its economic and employment value compared to other cities' economic conditions. This led to the extensive transformation of various land uses into built-up regions. The agricultural land and waterbodies class of was estimated to decrease from 56.00 km^2 to 44.36 km^2 and 21.34 km^2 – 10.85 km^2 , respectively, from 2010 to 2021. The change in the area for LULC in Dhaka City between 2010 and 2021 is mentioned in Table 4. The Built-up area increased by 34.986 km^2 (11%), while Waterbody, Vegetation, and Agricultural Land decreased by 10.49 km^2 (1%), 56.556 km^2 (8%), and 11.646 km^2 (3%), respectively. On the other hand, bare land increased by 43.707 km^2 (15%).

Different conditioning factors are required to simulate or predict future LULC patterns and discern the pattern of LULC change. Therefore, to model the potential change in LULC in Dhaka City, we utilized five factors that could influence such development (Fig. 5). For instance, the SRTM DEM was employed to calculate slope and elevation. The critical topographic elements of LULC change are elevation, slope, and aspect. The Euclidean distance approach was used to derive the distance from a road or river from the LULC maps. LULC is positively impacted by

proximity to a road and a river. However, LULCC decreases with increasing distance from rivers.

From 2010 to 2021, the most significant change in the LULC pattern was observed in the conversion of Agricultural Land to Built-up areas. The growth of built-up areas and barren land was most noticeable in Dhaka's western and northern parts (Fig. 6). From 2021 to 2030, the conversion of the Built-up area to other LULC types showed the most change in LULC patterns. Conversely, Dhaka City is expected to experience minimal and maximum area changes between 2030 and 2040. Between 2010 and 2040, practically all undeveloped land will be utilized for other LULCs. This is because LULC underwent significant transformations between 2010 and 2040. Urbanization increased the Built-up area of bare land, a common trend in this century.

Dhaka City, one of the world's fastest-growing cities, has experienced substantial changes in LULC classes in recent years. In this context, Fig. 7 illustrates the LULC transition scenario in Dhaka City during the study period. Between 2010 and 2021, the vegetation coverage underwent the most significant transition, with approximately 33.73 km^2 of vegetation being converted into built-up areas. From 2010 to 2040, around 29.14 km^2 of agricultural land was transformed into vegetation cover. From 2021 to 2030, approximately 16.577 km^2 of Agricultural Land was converted into vegetation; a similar scenario was observed for the years 2030 and 2040, when 23.10 km^2 of agricultural land was transformed into the vegetation class (see Fig. 8).

The LULC study conducted for the year 2030 anticipates that Waterbody, Vegetation, and Agricultural Land will persistently expand, particularly in Dhaka's northern, eastern, and southeastern regions. In the foreseeable future, agricultural land, water bodies, and vegetation are projected to continue their growth. Conversely, bare land is expected to decrease, as indicated by the area differences between each LULC

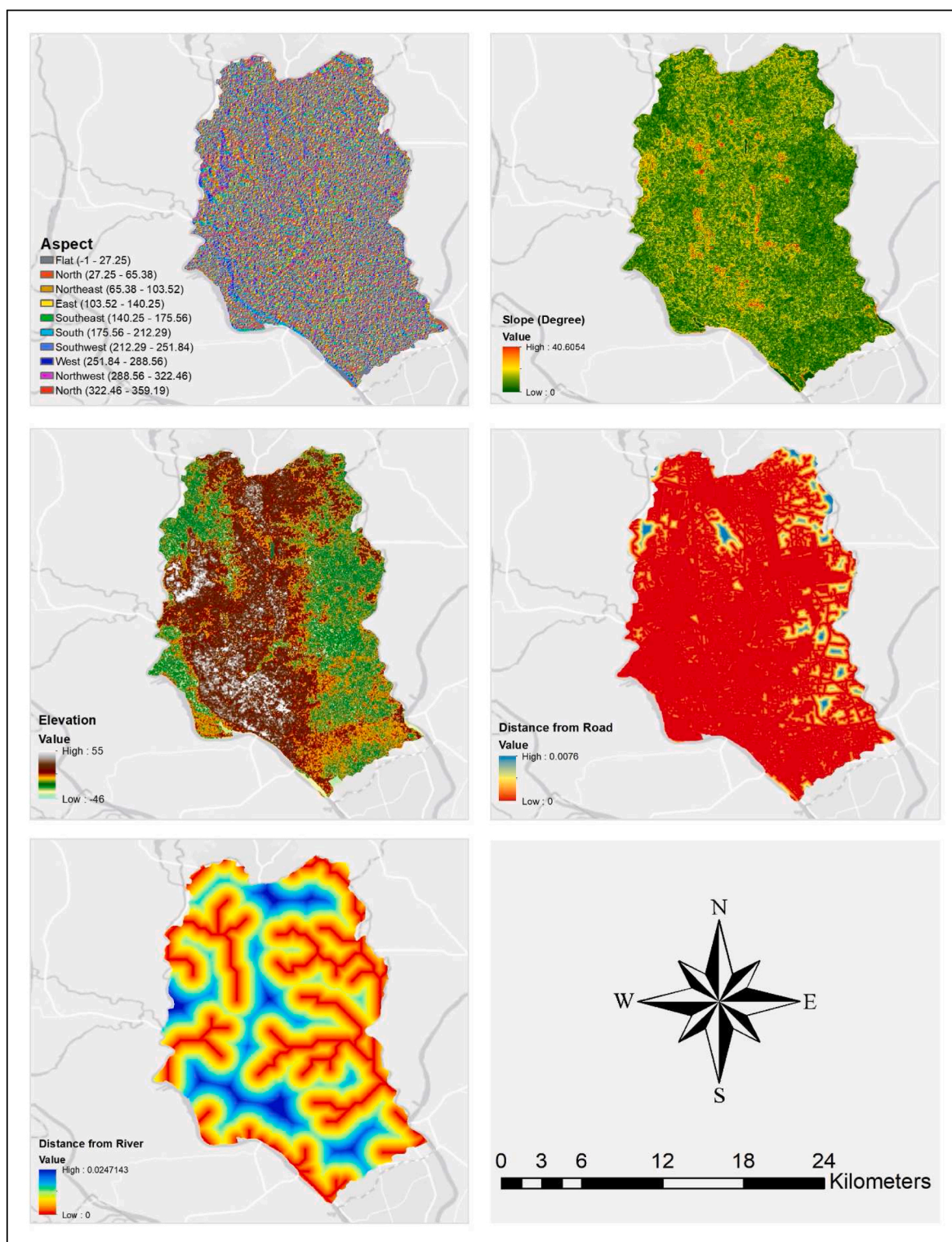


Fig. 5. LULC influencing Parameter.

class in the simulated LULC. By 2030 and 2040, the built-up area is estimated to be approximately 101.96 km² (33.91% of the total area) and 95.641 km² (31.81% of the total area), respectively. A marginal increase in water body area is expected between 2030 and 2040 (Table 5). Despite the potential for the growth rate to decelerate after 2021, Agricultural Land is projected to continue expanding until 2040. It is anticipated that vegetation will cover nearly all of the open area in the eastern part of Dhaka. On the other hand, the predicted LULC pattern in

the central region of Dhaka City is not expected to undergo significant changes.

The Kappa coefficient, as displayed in Table 6, was utilized to evaluate the performance of the LULC classifications in this study. Research has shown that for a comprehensive understanding of the categorized LULC maps derived from satellite observations, the overall accuracy should consistently exceed 70%. An examination of the accuracy of the categorized LULC maps revealed that the highest overall accuracy was

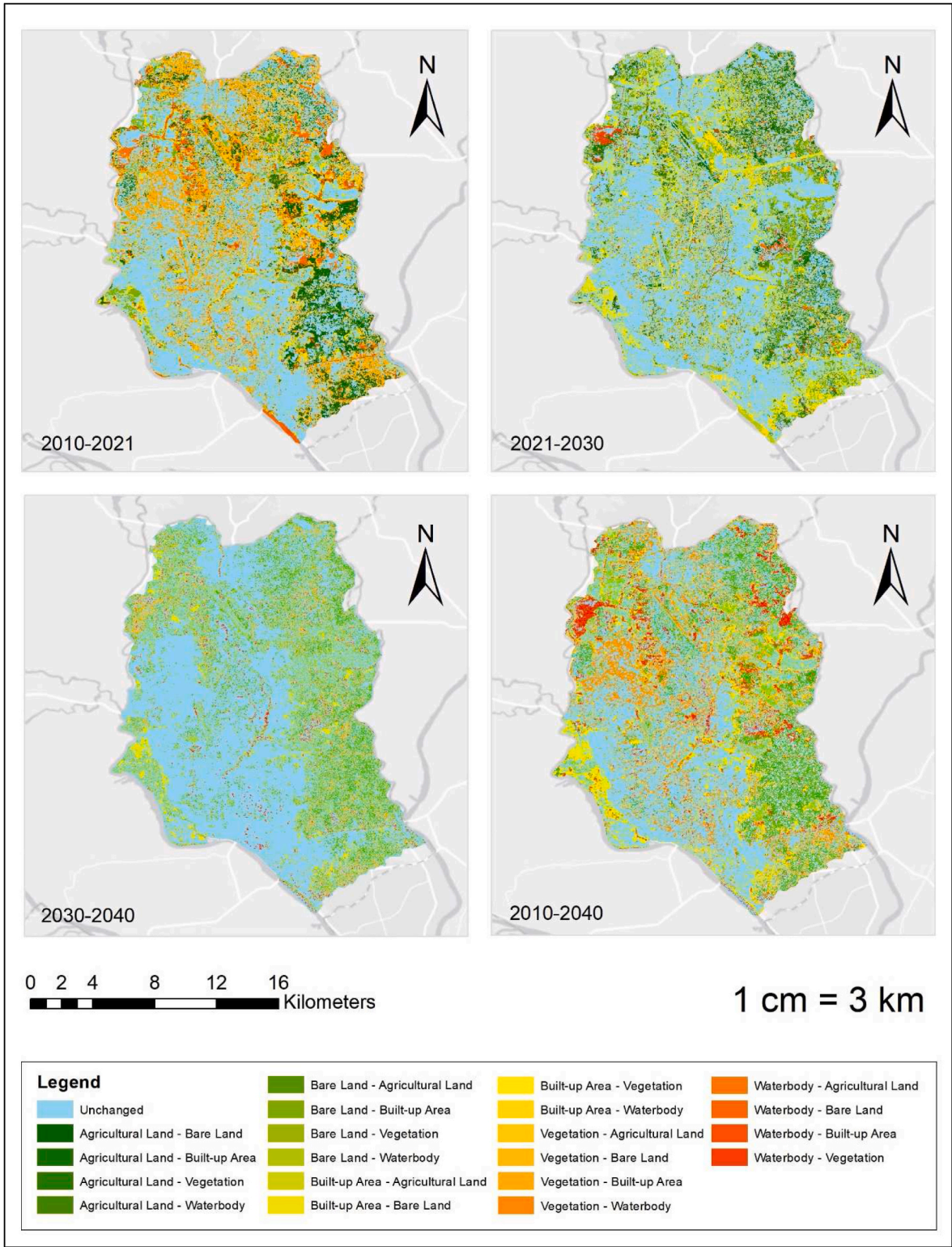


Fig. 6. Conversion of different LULC categories.

96% for 2021, with a corresponding Kappa value of 0.94. This was followed by 91% in 2015, with a corresponding Kappa coefficient of 0.88. The lowest accuracy was 89% for 2010, with a corresponding Kappa coefficient of 0.86. In general, the simulated LULC was 80% accurate for 2030, similar to the findings of another study, and 69% accurate for 2040, with Kappa coefficients of 0.74 and 0.59, respectively. The accuracy of the LULC for the year 2040 is based on the LULC data for

2021, which explains why the overall accuracy for 2040 is 69%. High accuracy and validation are significant aspects of LULC change detection, as found in this research. In summary, the CA-ANN model supports land management by providing accurate predictions, modeling complex relationships, managing extensive data, offering flexibility, and enhancing the reliability of land use and land cover change forecasts. This can assist land managers in their work.

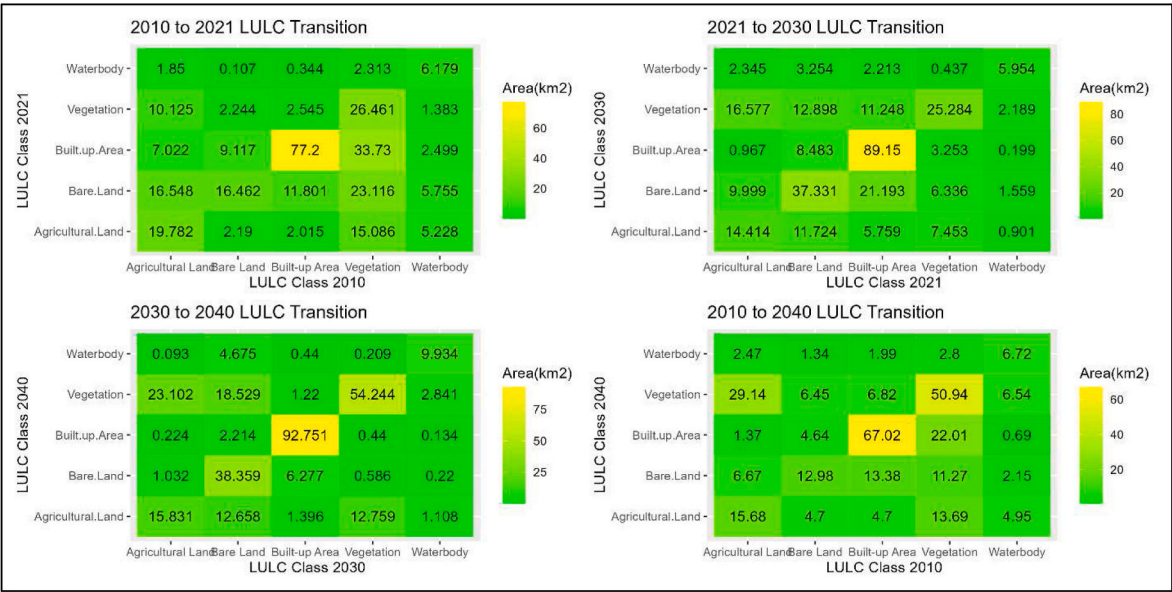


Fig. 7. LULC transition scenario at Dhaka City.

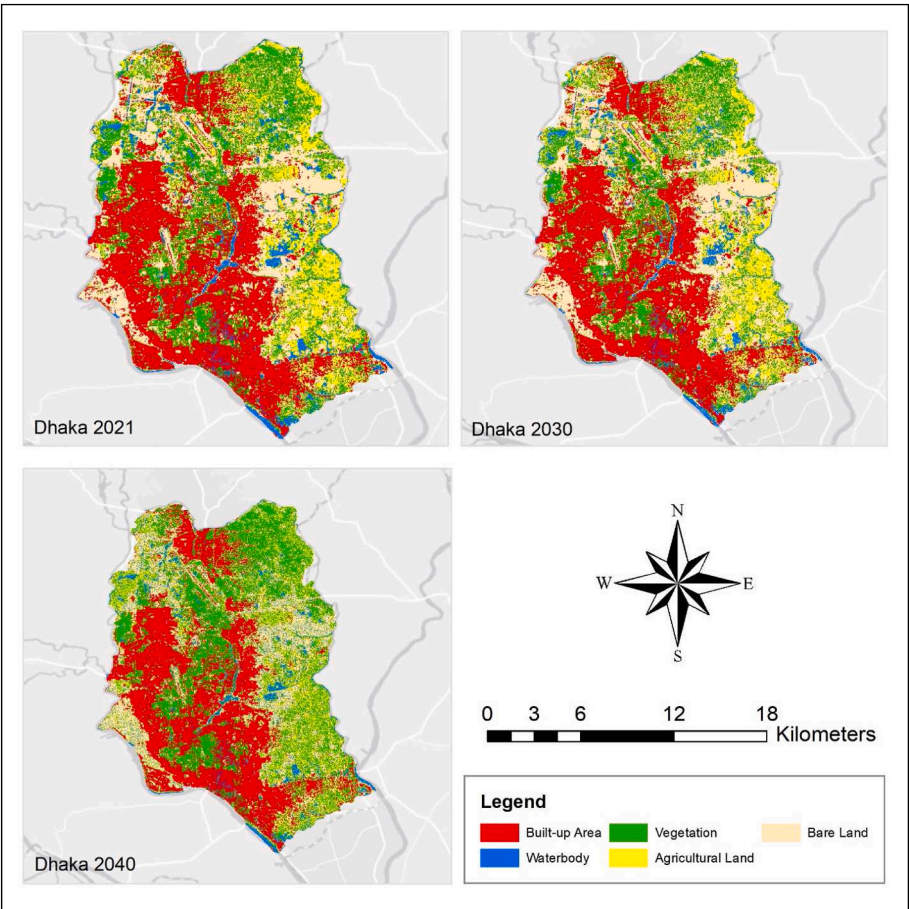


Fig. 8. Simulated LULC map of Dhaka city.

3.2. Estimation and prediction of ESVs from 2010 to 2040

In this section, the changes in ES were calculated for each LULC class by increasing the predicted area shifts by the ESV coefficients provided by Akber et al. (2018), as shown in Table 7. The results indicate that the

total expected ESV for Dhaka in 2010, 2015, 2021, 2030, and 2040 was 172.66, 158.37, 146.22, 144.23, and 160.23 million USD, respectively. During this period, built-up areas contributed the most to the total ESV, followed by vegetation and forest cover, water bodies, and agricultural land. Conversely, bare land did not contribute to the ESV. Although

Table 5

LULC area changes of Dhaka City from 2021 to 2040.

LULC Class	2021		Simulation 2030		Simulation 2040	
	Area (Km ²)	Area (%)	Area (Km ²)	Area (%)	Area (Km ²)	Area (%)
Built-up Area	127.514	42	101.965	33.91	95.641	31.81
Waterbody	10.850	4	14.117	4.7	15.231	5.07
Vegetation	43.199	14	68.112	22.65	99.815	33.2
Agricultural Land	44.361	15	40.162	13.36	43.632	14.51
Bare Land	74.758	25	76.325	25.38	46.364	15.42
Total	300.68	100	300.68	100	300.68	100

other studies found that water bodies are the greatest contributor to ESV for Dhaka and agriculture for Sundarbans, the LULC change in this study resulted in a net loss of US\$ 12.48 billion between 2010 and 2040.

Although the total ES value has progressively decreased over the past 11 years (2010–2021) and is expected to continue to do so in the future, the rate of decrease of ES has remained constant. According to the research, Dhaka's ESV trend suggests a decreasing process until 2030, after which the ESV will show a slight rising trend. During the research period, there was a gradual decline in ESV for agricultural land, water bodies, forest, and vegetation cover, with a rapid fall in ESV from 2010 to 2015 and 2015 to 2021. Water bodies and agricultural land experienced the most significant decline in ESV (US\$7.64 billion) between 2010 and 2040. Although the size of the built-up category expanded significantly over this period, the related growth in the built-up area ESV appeared to have a significant net influence on the total value of ES within the study area. In addition, the ESV of the forest and vegetation category changed slightly. The net effect of the bare land class did not affect the overall ES in the study area. Additionally, it can be shown that the yearly change in ESV from forest and vegetation land use (4.93%) increased between 2010 and 2015. In contrast, water bodies (584%) and forest and vegetation (7.08%) increased between 2015 and 2021, following the result of Akber et al. (2018).

The results of estimating the overall value of ES over five years using various value coefficients are shown in Table 8. A sensitivity evaluation was executed to estimate the accuracy of the findings. Using a change of around 50% from the initial value coefficient provided by Costanza et al. (2014), the Sensitivity Coefficient (CS) was determined. To be precise, each land use category's estimated coefficient service value must be < 1 for the estimated ESV. The ES value coefficient was excellent if the value of CS is consistently <1, proving that the estimate of ESV was typically inelastic concerning the coefficient value. Our study's sensitivity coefficient was lower than one, indicating that the overall value of ES was calculated without regard for the coefficient value.

The overall value of ES was slightly affected by water bodies in the year 2010 adjustment of the value coefficient of agriculture, changing by less than 10% for $\pm 50\%$ of the value coefficient, compared to Built-up area and vegetation cover, which had a significant influence on ES, changing by more than 15% for 50% of the value coefficient. In respect, the effect of built-up area on ES increases from 17.85, 22.21, and 29.04

in 2010, 2015, and 2021, respectively, to 23.55 and 19.04 in 2030 and 2040, respectively. Additionally, the effects of agriculture and water bodies have a progressive 50% value coefficient drop on the overall value of ES over time. Additionally, when the value coefficient for these LULC groups was decreased by 50%, the CS value heights for built-up areas ranged from 0.36 to 0.58, while the lowest values for other LULC types were 0.09 for water bodies, 0.16 for forests and vegetation, and 0.15 for agricultural land. In ES, the barren terrain had neither a positive nor a negative effect. Moreover, this finding indicates that the ES value estimations were stable over time, even though they were significantly undervalued or inflated.

4. Discussion

This study provides important insights into the changes in LULC of Dhaka City over the past few decades and their impacts on ESV. The analysis of satellite imagery from 2010 to 2021 reveals concerning trends of built-up area expansion at the expense of natural habitats like waterbodies, vegetation, and agricultural land. However, the predicted decrease in built-up areas coupled with the recovery of other LULC types by 2030–2040 is encouraging. It indicates that amidst rapid urbanization, efforts are being made towards balanced and sustainable urban development.

The predominant loss of vegetation, wetlands, and cropland to ever-increasing concrete jungles has become a ubiquitous trait of the world's megacities. Dhaka is no exception. Overpopulation, industrialization, and economic activities have fueled uncontrolled urban sprawl, suffocating the city's ecosystems (Dewan and Yamaguchi, 2009). The 18% increase in built-up land from 2010 to 2021 is a testament to the unbridled growth of residential, commercial, and industrial infrastructure. This perpendicular expansion has annexed wetlands and agricultural fields that once supported the city's rich biodiversity and provided ES to its inhabitants (Burkhard et al., 2009). For instance, the destruction of lakes, canals, and seasonal floodplains for construction has led to the loss of natural drainage, frequent urban flooding, and waterlogging. The decline in cropland has also reduced food provisioning services (Benis and Ferrão, 2017). However, the projected decline in the built-up area after 2021 indicates that the city could move towards a more balanced approach. There appears to be greater collective awareness that blind urbanization jeopardizes ecosystems integral to human well-being and survival (McPhearson et al., 2015).

The total ESV for Dhaka shows an overall declining trend from 2010 to 2040. This is unsurprising given the predominant degradation of ecosystems. Other major cities with rampant development like Beijing, Shanghai and Guangzhou, have also reported massive declines in ESV over the past few decades (Kong et al., 2014). The economic valuation provides a quantitative estimate of the costs of unsustainable urbanization to society, the environment, and future generations (Ninan and Inoue, 2013). For Dhaka, the loss of wetlands and agricultural land contributed most to the decline of ESV. Wetlands are among the most threatened habitats globally, with 64% lost in the 20th century (Zedler and Kercher, 2005). Dhaka's wetlands provide invaluable services,

Table 6

Accuracy of LULC map.

LULC Class	2010		2015		2021		Simulation 2030		Simulation 2040	
	UA	PA	UA	PA	UA	PA	UA	PA	UA	PA
Built-up Area	1	1	0.92	0.85	0.97	0.99	0.96	0.86	0.97	0.75
Waterbody	0.96	0.91	1	1	1	1	0.81	0.80	0.74	0.72
Vegetation	0.79	0.86	0.93	0.89	0.91	0.97	0.83	0.92	0.58	0.99
Agricultural Land	0.88	0.84	0.71	0.83	0.94	0.83	0.55	0.72	0.35	0.45
Bare Land	0.96	1	0.81	0.87	0.98	0.96	0.66	0.65	0.58	0.43
Overall Accuracy (%)	0.89		0.91		0.96		0.80		0.69	
Kapp Coefficient	0.86		0.88		0.94		0.74		0.59	

*UA= User's accuracy (%), PA= Producer's accuracy (%).

Table 7
Individual ESV for each year from 2010 to 2040 (*10⁻⁶).

LULC Type	Individual ESV Changes					ESV Changes				Annual Rate of ESV changes			
	2010	2015	2021	2030	2040	2010–2015	2015–2021	2021–2030	2030–2040	2010–2015	2015–2021	2021–2030	2030–2040
Built-up area	61.63	70.35	84.94	67.92	63.71	8.72	14.59	−17.02	−4.21	2.83	3.46	−2.23	−0.62
Waterbody	26.70	20.91	13.57	17.66	19.06	−5.79	−7.33	4.09	1.39	−4.34	−5.84	3.35	0.79
Forest and Vegetation	53.15	40.04	23.02	36.29	53.18	−13.11	−17.02	13.27	16.89	−4.93	−7.09	6.41	4.65
Agricultural land	31.18	27.08	24.70	22.36	24.29	−4.10	−2.38	−2.34	1.93	−2.63	−1.47	−1.05	0.86
Bare land	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total	172.66253	158.37261	146.22428	144.23047	160.23427								

including groundwater recharge, water purification, flood control, and microclimate regulation (Gupta et al., 2023). Their wanton destruction has stripped the city of its natural safety net against climate change vagaries like erratic rainfall, storm surges and heatwaves (Dasgupta et al., 2018). Meanwhile, the loss of arable land shrinks Dhaka's capacity for local food production, making it more dependent on external sources (Benis and Ferrão, 2017). The declining ESV underscores the environmental fragility induced by Dhaka's uncontrolled expansion. It provides economic incentives for policymakers to focus on ecosystem conservation and wise management of natural resources (Alauddin and Quiggin, 2008).

This study has some notable strengths. Using satellite remote sensing enabled a spatial and temporal analysis of LULC changes based on real data, providing an objective picture of on-ground transformations. Integrating advanced machine learning algorithms strengthened the accuracy and reliability of LULC classification and future predictions. The valuation of ESV based on LULC dynamics provides a quantitative, economic perspective of ecosystem changes, complementing the spatial analysis. However, certain limitations must be acknowledged. Firstly, the classification was based solely on LULC types without considering ecosystem quality, species composition or biodiversity (Rahman and Szabó, 2021). Despite no net change in area, land use intensification and degradation can reduce ESV. Secondly, the study provides a broad estimate of ESV using global values but does not account for location-specific variations (Zhang et al., 2022). Valuation based on field studies could yield more accurate, localized estimates (Nelson et al., 2009). Lastly, the focus on a single city limits the generalizability of the results. Comparative analyses across different urban regions could offer deeper insights into the nexus between urbanization patterns, ecosystems, and ESV.

Nonetheless, this study makes a significant contribution by unraveling LULC transformations in Dhaka City and examining their environmental and economic impacts through an ES lens. The results call for urgent strategies to safeguard Dhaka's remaining natural habitats and restore degraded ecosystems to enhance their functionality. Policy initiatives must focus on controlled urban expansion through sustainable land use planning by establishing protected wetlands and agricultural zones. Stricter environmental regulations are needed to prevent commercial encroachment on water bodies and greenery. Investments in ecological restoration and green infrastructure development, such as wetland revitalization or urban forestry, could effectively enhance ESV. Furthermore, community participation must be promoted in urban ecological management by incentivizing local stewardship of green spaces (Andersson et al., 2014). The study provides an information base to guide policymaking for balanced urban development that harmonizes gray and green infrastructure. It highlights that the path to sustainable, resilient, and livable cities lies in conserving and nurturing the ecosystems underpinning human survival.

The projected recovery of wetlands, vegetation and cropland by 2030–2040 is a positive sign and an opportunity to reverse the damage. It indicates evolving societal perspectives on urban sustainability and improving environmental awareness. However, translating this recovery into enhanced ESV depends on the effectiveness of conservation and restoration efforts. Legal protection of wetlands and agricultural areas is essential but not sufficient. Stricter monitoring and enforcement against ecosystem degradation are needed. Restored wetlands and green areas should mimic natural structures using native species. Sustainable agricultural practices like organic farming could enhance soil health and food production capacity. Community-driven initiatives, partnerships, and outreach programs can generate public support for ecosystem management. Overall, a holistic approach is required not just to protect existing natural habitats but actively invest in their revitalization for maximizing ESV.

The study provides a framework for connecting LULC patterns with ESV that could be applied in other urban contexts. With some modifications, it can help assess ecosystem changes and economic impacts

Table 8
Changes in total estimated ESV and CS.

LULC	2010		2015		2021		2030		2040	
	ESV%	CS	ESV%	CS	ESV%	CS	ESV%	CS	ESV%	CS
Built-up area $\pm 50\%$	17.85	0.36	22.21	0.44	29.04	0.58	23.55	0.47	19.88	0.40
Water body $\pm 50\%$	7.73	0.16	6.60	0.13	4.64	0.09	6.12	0.12	5.95	0.12
Forest and Vegetation $\pm 50\%$	15.39	0.31	12.64	0.25	7.87	0.16	12.58	0.25	16.60	0.33
Agricultural land $\pm 50\%$	9.03	0.18	8.55	0.17	8.44	0.17	7.75	0.16	7.58	0.15
Bare land $\pm 50\%$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

across different cities. However, future research could build on this study in several ways. Longer time-scale analyses over 50–100 years could reveal macro-level LULC and ESV shifts. Comparative studies across rapidly developing Asian cities could identify region-specific trends, common challenges, and transferable solutions. Exploring the effectiveness of emerging ecosystem management policies using scenario modeling will offer practical insights for decision-makers. Qualitative surveys and social data could provide a deeper understanding of human dimensions like community perceptions of ES. An interdisciplinary systems approach combining spatiotemporal, economic, and social perspectives is needed to guide pathways for resilient urban development. This study provides an important foundation to take forward that mission.

Examining LULC changes and the decline in ESV presents a worrying image of Dhaka's environmental well-being under haphazard urbanization. This underscores the urgent need for collective action to rejuvenate the city's lost ecosystems and preserve its remaining natural capital before it descends into an ecological crisis. The silver lining is that there is still room for improvement if we can secure institutional commitment and raise public awareness towards sustainable development. The opportunity has not yet passed, and the solution lies in understanding that gray infrastructure must coexist with green. Dhaka has the potential to regain its status as a lively water city, teeming with wetlands and green spaces, by balancing urban expansion with ecosystem preservation. The effectiveness of such initiatives will be evident in a reversal of the ESV decline. More importantly, it will enhance the quality of life for residents through improved air quality, cooler microclimates, flood control, and access to local food sources. This study offers crucial insights to guide policies that restore Dhaka's urban ecosystems and direct the city towards a sustainable future.

5. Conclusion

This research highlights concerning trends in LULC changes in Dhaka City over the past decade, characterized by uncontrolled expansion of built-up areas at the cost of vital ecosystems. However, the predicted stabilization and recovery of natural habitats by 2030–2040 provide hope for a more sustainable future through prudent planning and management. The declining ESV quantified in this study underscores Dhaka's ecosystems' fragility and diminished capacity to support urban communities. Yet, they also present an economic case for urgent policy action to reverse the damage before it becomes irreversible.

The predominant loss of wetlands, vegetation, and cropland to ever-increasing concrete jungles has become a ubiquitous trait of rapidly developing cities worldwide. Global analyses have revealed massive declines in natural habitats and ES in major urban centers over the past century. Dhaka is no exception to this unsustainable growth pattern, as evidenced by the significant expansion of built-up land between 2010 and 2021, annexing vital green and blue infrastructure. Unplanned urbanization has not just suffocated biodiversity but also stripped ecosystems of their ability to provide human health and well-being services. The social costs of this ecological degradation are severe, from increasing flood severity and waterlogging due to wetland destruction to heightened air pollution and urban heat island effects resulting from the loss of greenery. Yet amidst the doom, Dhaka seems to be progressing

towards a more balanced approach to development. The predicted decline in built-up area accompanied by recovery of wetlands, vegetation and cropland after 2021 indicates evolving perspectives that blind urbanization jeopardizes ecosystems integral to resilience and sustainability. The declining ESV, especially driven by loss of wetlands and agricultural land, provides a quantitative estimate of how unchecked growth is diminishing Dhaka's natural capital assets and environmental revenue. Ecosystem valuation has gained increasing traction globally as a tool to inform policy and decision-making. For Dhaka, the substantial ESV declines projected from 2010 to 2040 make a compelling economic case for urgent strategies to conserve, restore, and wisely manage remaining ecosystems. The fact that wetlands and cropland contributed most to the dropping ESV is unsurprising, given their extensive degradation over the years. Yet, it highlights where intervention efforts should be targeted and investments prioritized. The good news is that the projected recovery of these LULC types by 2030–2040 offers opportunities to protect existing ecosystems and actively restore their functionality to recoup the lost ESV. Conversely, despite their predominant growth, the minimal net change in ESV from built-up areas spotlights the need to balance urban development with green infrastructure through ecological design.

This timely study makes a significant contribution by unraveling LULC dynamics in Dhaka City using advanced spatiotemporal techniques and connecting the transformations to their environmental and economic impacts through an ES perspective. The spatial analysis based on satellite remote sensing and machine learning algorithms provided an objective picture of on-ground changes. Integrating LULC data with ecosystem service valuation enabled a more holistic and quantitative assessment of how urbanization patterns influence ecological infrastructure and society.

This research sends an unequivocal message that the unabated expansion of gray spaces at the expense of green and blue natural capital precipitates Dhaka towards an ecological cliff-edge. It highlights the urgent need for collective action to restore damaged ecosystems and nurture those that remain before the city slips into an environmental abyss. The window of opportunity still exists to reverse the trends, but it is closing fast. Forward-looking policies and public awareness are vital to fostering a culture of sustainable development that harmonizes urban growth with ecological conservation. This will ensure Dhaka can reclaim its identity as a liveable city integrated with thriving wetlands, waterways, and greenery. The success of such efforts will be reflected in a rebound of ESV over the coming decades. More importantly, it will enhance the quality of life for citizens through cooler microclimates, cleaner air and water, flood regulation, and disaster resilience. In essence, restoring Dhaka's lost ecosystems translates to securing its own sustainable future.

While this study provides vital baseline insights, further research is needed to support informed decision-making on ecosystem management and sustainable urban development in Dhaka City. Some potential avenues include Long-term analyses of LULC and ESV changes over 50–100 years using historical remote sensing data and field inventories. This can reveal macro-level shifts beyond the study's limited time scale. In addition, a comparative assessment of LULC-ESV links across rapidly urbanizing cities in South Asia can be conducted to identify common patterns, challenges, and transferable solutions. Overall, a

multidimensional systems approach combining spatiotemporal, economic, and social perspectives is vital to paint a fuller picture. Integrative models and mixed methods can unravel the complex couplings between urbanization patterns, ecosystem transformations, and human well-being. This will provide a robust foundation to guide Dhaka's trajectory toward resilience and sustainability.

Funding information

The author(s) received no financial support for the research, authorship, and/or publication of this article.

Ethical approval

Not applicable.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Availability of data and materials

Data will send based on request.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Sujit Kumar Roy: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Md. Tauhidul Alam:** Conceptualization, Data curation, Formal analysis, Methodology, Resources, Software, Visualization, Writing – original draft, Writing – review & editing. **Pratik Mojumder:** Conceptualization, Data curation, Formal analysis, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Ismail Mondal:** Conceptualization, Supervision, Validation, Writing – original draft, Writing – review & editing. **Abdulla - Al Kafy:** Conceptualization, Formal analysis, Investigation, Methodology, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Mithun Dutta:** Formal analysis, Resources, Software, Validation, Visualization, Writing – original draft. **Md Nahid Ferdous:** Data curation, Formal analysis, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Md. Abdullah Al Mamun:** Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Sania Binte Mahtab:** Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

Not Applicable

References

- Akber, M.A., Khan, M.W.R., Islam, M.A., Rahman, M.M., Rahman, M.R., 2018. Impact of land use change on ecosystem services of southwest coastal Bangladesh. *J. Land Use Sci.* 13 (3), 238–250. <https://doi.org/10.1080/1747423X.2018.1529832>.
- Alauddin, M., Quiggin, J., 2008. Agricultural intensification, irrigation and the environment in South Asia: issues and policy options. *Ecol. Econ.* 65 (1), 111–124. <https://doi.org/10.1016/j.ecolecon.2007.06.004>.
- Andersson, E., et al., 2014. Reconnecting cities to the biosphere: stewardship of green infrastructure and urban ecosystem services. *Ambio* 43 (4), 445–453. <https://doi.org/10.1007/s13280-014-0506-y>.
- Aneesa Satya, B., Shashi, M., Deva, P., 2020. Future land use land cover scenario simulation using open source GIS for the city of Warangal, Telangana, India. *Appl. Geomatics* 12 (3), 281–290. <https://doi.org/10.1007/s12518-020-00298-4>.
- Baig, M.F., Mustafa, M.R.U., Baig, I., Takaijudin, H.B., Zeshan, M.T., 2022. Assessment of land use land cover changes and future predictions using CA-ANN simulation for selangor, Malaysia. *Water* 14 (3), 402. <https://doi.org/10.3390/W14030402>, 2022, Vol. 14, Page 402.
- Benis, K., Ferrão, P., 2017. Potential mitigation of the environmental impacts of food systems through urban and peri-urban agriculture (UPA) – a life cycle assessment approach. *J. Clean. Prod.* 140, 784–795. <https://doi.org/10.1016/j.jclepro.2016.05.176>.
- Birhane, E., et al., 2019. Land use land cover changes along topographic gradients in Hugumburda national forest priority area, Northern Ethiopia. *Remote Sens. Appl. Soc. Environ.* 13, 61–68. <https://doi.org/10.1016/j.rsase.2018.10.017>.
- Burkhard, B., Kroll, F., Müller, F., Windhorst, W., 2009. Landscapes' capacities to provide ecosystem services - a concept for land-cover based assessments. *Landsc. Online* 15 (1), 1–22. <https://doi.org/10.3097/LO.200915>.
- Chen, Y., Li, X., Liu, X., Ai, B., Li, S., 2016. Capturing the varying effects of driving forces over time for the simulation of urban growth by using survival analysis and cellular automata. *Landsc. Urban Plann.* 152, 59–71. <https://doi.org/10.1016/j.landurbplan.2016.03.011>.
- Costanza, R., et al., 1997. The value of the world's ecosystem services and natural capital. *Nat* 387 (6630), 253–260. <https://doi.org/10.1038/387253a0>, 1997 3876630.
- Dasgupta, S., Hossain, M.M., Huq, M., Wheeler, D., 2018. Climate change, salinization and high-yield rice production in coastal Bangladesh. *Agric. Resour. Econ. Rev.* 47 (1), 66–89. <https://doi.org/10.1017/age.2017.14>.
- Dewan, A.M., Yamaguchi, Y., 2009. Land use and land cover change in Greater Dhaka, Bangladesh: using remote sensing to promote sustainable urbanization. *Appl. Geogr.* 29 (3), 390–401. <https://doi.org/10.1016/j.apgeog.2008.12.005>.
- Gupta, G., Shrivastava, R., Khan, J., Singh, N.K., 2023. Emerging approaches for sustainable urban metabolism. *Urban Metab. Clim. Chang.* 247–273. https://doi.org/10.1007/978-3-031-29422-8_13.
- Khwarahm, N.R., 2021. Spatial modeling of land use and land cover change in Sulaimani, Iraq, using multitemporal satellite data. *Environ. Monit. Assess.* 193 (3) <https://doi.org/10.1007/s10661-021-08959-6>.
- Khwarahm, N.R., Najmaddin, P.M., Ararat, K., Qader, S., 2021. Past and future prediction of land cover land use change based on earth observation data by the CA-Markov model: a case study from Duhok governorate, Iraq. *Arabian J. Geosci.* 14 (15) <https://doi.org/10.1007/s12517-021-07984-6>.
- Kong, F., Yin, H., James, P., Hutyra, L.R., He, H.S., 2014. Effects of spatial pattern of greenspace on urban cooling in a large metropolitan area of eastern China. *Landsc. Urban Plann.* 128, 35–47. <https://doi.org/10.1016/j.landurbplan.2014.04.018>.
- Kreuter, U.P., Harris, H.G., Matlock, M.D., Lacey, R.E., 2001. Change in ecosystem service values in the san antonio area, Texas. *Ecol. Econ.* 39 (3), 333–346. [https://doi.org/10.1016/S0921-8009\(01\)00250-6](https://doi.org/10.1016/S0921-8009(01)00250-6).
- Li, T., Li, W., 2015. Multiple land use change simulation with Monte Carlo approach and CA-ANN model, a case study in Shenzhen, China. *Environ. Syst. Res.* 4 (1), 1–10. <https://doi.org/10.1186/s40068-014-0026-6>.
- McPhearson, T., Andersson, E., Elmqvist, T., Frantzeskaki, N., 2015. Resilience of and through urban ecosystem services. *Ecosyst. Serv.* 12, 152–156. <https://doi.org/10.1016/J.ECOSER.2014.07.012>.
- Nahrin, K., 2020. Environmental area conservation through urban planning: case study in Dhaka. *J. Prop. Plan. Environ. Law* 12 (1), 55–71. <https://doi.org/10.1108/JPPPEL-11-2018-0033>.
- Naikoo, M.W., Rihan, M., Ishtiaque, M., Shahfahad, 2020. Analyses of land use land cover (LULC) change and built-up expansion in the suburb of a metropolitan city: spatio-temporal analysis of Delhi NCR using landsat datasets. *J. Urban Manag.* 9 (3), 347–359. <https://doi.org/10.1016/j.jum.2020.05.004>.
- Nelson, E., et al., 2009. Modeling multiple ecosystem services, biodiversity conservation, commodity production, and tradeoffs at landscape scales. *Front. Ecol. Environ.* 7 (1), 4–11. <https://doi.org/10.1890/080023>.
- Ninan, K.N., Inoue, M., 2013. Valuing forest ecosystem services: case study of a forest reserve in Japan. *Ecosyst. Serv.* 5, 78–87. <https://doi.org/10.1016/j.ecoser.2013.02.006>.
- Ongsomwang, S., Pattanakiat, S., Srisuwan, A., 2019. Impact of land use and land cover change on ecosystem service values: a case study of khon kaen city, Thailand. *Environ. Nat. Resour. J.* 17 (4), 43–58. <https://doi.org/10.32526/ENNRJ.17.4.2019.30>.

- Pandey, S., Kumari, N., 2023. Prediction and monitoring of LULC shift using cellular automata-artificial neural network in Jumar watershed of Ranchi District, Jharkhand. *Environ. Monit. Assess.* 195 (1) <https://doi.org/10.1007/s10661-022-10623-6>.
- Past, A., Land, F., Land, U., Kumar, P., 2022. Cellular automata-based artificial neural network model for cover dynamics. *Agronomy* 12, 2772.
- Pullanikkatil, D., Palamuleni, L.G., Ruhiiga, T.M., 2016. Land use/land cover change and implications for ecosystems services in the Likangala River Catchment, Malawi. *Phys. Chem. Earth* 93, 96–103. <https://doi.org/10.1016/j.pce.2016.03.002>.
- Rahaman, M.A., Kalam, A., Al-Mamun, M., 2023. Unplanned urbanization and health risks of Dhaka City in Bangladesh: uncovering the associations between urban environment and public health. *Front. Public Health* 11, 1269362. <https://doi.org/10.3389/fpubh.2023.1269362/BIBTEX>.
- Rahman, F., Rahman, M.T.U., 2023. Use of cellular automata-based artificial neural networks for detection and prediction of land use changes in North-Western Dhaka City. *Environ. Sci. Pollut. Res.* 30 (1), 1428–1450. <https://doi.org/10.1007/s11356-022-22079-9>.
- Rahman, M.M., Szabó, G., 2021. Impact of land use and land cover changes on urban ecosystem service value in Dhaka, Bangladesh. *L.* 2021, Vol. 10, Page 793, 10 (8), 793. <https://doi.org/10.3390/LAND10080793>.
- Rahman, M.H., Khan, T.Z., Raju, R.A., Huq, M.J., 2019. Urban eco-sustainability and disaster risk reduction by implementing vertical garden in Dhaka city eco-system services and urban disaster risk management. *Proc. Int. Conf. Disaster Risk Manag.* 206–213.
- Saha, T.K., Pal, S., Sarkar, R., 2021. Prediction of wetland area and depth using linear regression model and artificial neural network based cellular automata. *Ecol. Inf.* 62 (October 2020), 101272 <https://doi.org/10.1016/j.ecoinf.2021.101272>.
- Sanderson, R., 2010. Introduction to Remote Sensing.
- Saputra, M.H., Lee, H.S., 2019. Prediction of land use and land cover changes for North Sumatra, Indonesia, using an artificial-neural-network-based cellular automaton. *Sustain. Times* 11 (11), 1–16. <https://doi.org/10.3390/su11113024>.
- Tolessa, T., Senbeta, F., Kidane, M., 2017. The impact of land use/land cover change on ecosystem services in the central highlands of Ethiopia. *Ecosyst. Serv.* 23 (June 2016), 47–54. <https://doi.org/10.1016/j.ecoser.2016.11.010>.
- Tomar, P., et al., 2021. Gis-based urban flood risk assessment and management—a case study of Delhi national capital territory (NCT), India. *Sustain. Times* 13 (22), 1–20. <https://doi.org/10.3390/su132212850>.
- Verma, P., Raghubanshi, A., Srivastava, P.K., Raghubanshi, A.S., 2020. Appraisal of kappa-based metrics and disagreement indices of accuracy assessment for parametric and nonparametric techniques used in LULC classification and change detection. *Model. Earth Syst. Environ.* 6 (2), 1045–1059. <https://doi.org/10.1007/s40808-020-00740-x>.
- Wu, F., 2002. Calibration of stochastic cellular automata: the application to rural-urban land conversions. *Int. J. Geogr. Inf. Sci.* 16 (8), 795–818. <https://doi.org/10.1080/13658810210157769>.
- Yin, H., Islam, M.S., Ju, M., 2021. Urban river pollution in the densely populated city of Dhaka, Bangladesh: big picture and rehabilitation experience from other developing countries. *J. Clean. Prod.* 321, 129040 <https://doi.org/10.1016/J.JCLEPRO.2021.129040>.
- Zedler, J.B., Kercher, S., 2005. Wetland resources: status, trends, ecosystem services, and restorability. *Annu. Rev. Environ. Resour.* 30, 39–74. <https://doi.org/10.1146/annurev.energy.30.050504.144248>.
- Zhang, X., Ren, W., Peng, H., 2022. Urban land use change simulation and spatial responses of ecosystem service value under multiple scenarios: a case study of Wuhan, China. *Ecol. Indic.* 144 (October), 109526 <https://doi.org/10.1016/j.ecolind.2022.109526>.
- Zinia, N.J., McShane, P., 2018. Ecosystem services management: an evaluation of green adaptations for urban development in Dhaka, Bangladesh. *Landsc. Urban Plann.* 173 (January), 23–32. <https://doi.org/10.1016/j.landurbplan.2018.01.008>.
- Molusce – quick and convenient analysis of land cover changes | NextGIS [Online]. Available: <https://nextgis.com/blog/molusce/>. (Accessed 14 November 2023).